

Track 4: The Induced Leak (Faraday's Law)

Inductive Crosstalk Data: Showing voltage flashes upon exact signal transitions due to changing magnetic flux.

Time (ms)	Column A (Source Cable)	Column B (Attacker Cable)	Physical State
t = 1	0 V	0 mV	Steady state
t = 2	0 V → 5 V	+800 mV (Flash & Return to 0)	Changing magnetic flux (Rising Edge)
t = 3	5 V	0 mV	Steady state
t = 4	5 V → 0 V	-800 mV (Flash & Return to 0)	Changing magnetic flux (Falling Edge)
t = 5	0 V	0 mV	Steady state

Explanation: Why does it flash and return to zero?

The Scientific Reason: According to Faraday's Law, a voltage is only induced in Cable B when the magnetic field is *actively changing*.

- When Cable A transitions from 0V to 5V, the current rapidly increases. This creates a growing magnetic field (**Changing Magnetic Flux**), which generates the sudden +800mV flash in Column B.
- However, once Cable A settles at 5V or 0V, the current becomes steady. A steady current creates a steady magnetic field. Because there is no longer a *change* in the flux, the induced voltage in Column B immediately drops back to 0 mV.

This is why the attacker only sees the **exact moment** of the transition, not the data itself!